

FAQ Analysis

1. What are the differences between Muto robot S1 version, S2 version, and RS version?

A: The S1 version uses PS2 controller control method. After successful installation, you can use the PS2 controller to control it without programming. The S2 version uses Jetson Nano or Raspberry Pi main controller and supports Python programming control. The RS version includes all functions of S2 and also contains the ROS2 system, offering more diverse gameplay.

2. How to handle when the camera cannot read data?

A: Confirm that the hardware connection is fine. Please exit the program, re-plug the device's USB cable, and check if there is a /dev/video0 device number. If not, please restart the system.

3. How is the robot powered?

A: The car comes with a battery pack from the factory. Connect the battery pack's T-type interface to the expansion board's battery interface through a converter cable, turn on the main power switch, and the expansion board integrates voltage conversion chips to provide power for various devices.

4. Which functions on the expansion board are managed by the microcontroller?

A: The parts managed by the microcontroller on the expansion board include: status indicator lights, buzzer, attitude sensor, RESET key, PWM servo interface, serial servo interface, etc.

5. Why do I need to close the APP control program? What impact does it have on program development?

A: For convenient experience of the control program, the robot automatically runs the APP control program when booting, but it will occupy resources such as cameras and serial ports. Before developing actual routines, you need to close the APP control program first to avoid errors when routines call resources such as cameras and serial ports. If you don't use APP control for a long time, you can permanently close the APP control program according to the tutorial.

6. What if one leg of the robot isn't touching the ground when it returns to a stationary position?

A: Please connect to the APP control software, find the robot calibration, and recalibrate the servo motors of the robot's legs according to the calibration steps.

7. What if the robot still prioritizes connecting to its home Wi-Fi signal after enabling hotspot mode?

A: Since both hotspot mode and Wi-Fi connection use Wi-Fi devices and have startup priorities, please set the priority of hotspot mode to be higher than your home Wi-Fi signal. This will ensure that the robot transmits hotspot signals first.

8. What if the color recognition or color tracking function is inaccurate?

A: Colors of objects may vary, and differences in the environment and lighting conditions can also cause variations. Please calibrate according to the HSV calibration tutorial and update the calibrated HSV data in the corresponding example.

9. What should I do if the robot automatically crouches down and emits a "beep beep beep" alarm?

A: This indicates the robot is in low voltage protection mode. Please turn off the robot and charge it promptly.

10. What are the meanings of the indicator lights on the robot's expansion board?

A: POWER indicator light: Indicates the robot's power supply status. Normally, it's constantly lit; off indicates an abnormality. MCU indicator light: Indicates the microcontroller's operating status. Normally, it flashes twice every 3 seconds. When the battery is low, it flashes continuously. A constantly lit or off indicator shows a microcontroller malfunction.

11. Why do I sometimes get errors when using the camera?

Starting the OpenCV program occupies the Video0 port. Starting the depth camera simultaneously can cause port contention. If the camera fails to start at this time, try unplugging and replugging the camera.

12. Why do customers who purchased a voice board encounter errors when starting a large program?

If the voice board is plugged into the port during the first startup, you'll find an error message indicating that the expansion board cannot be found. This is because both the voice board and the expansion board are recognized as "myserial". You need to unplug the voice board before starting. If you need to use the voice board, refer to the voice board port binding tutorial.

13. Why does starting the Astro_Pro_Plus depth camera separately in Docker result in an error?

On the host machine, please use the following command to check the camera's port:

```
ll /dev/astra*
```

The depth camera has two ports. If only one port is displayed, like this:

```
jetson@yahboom:~$ ll /dev/astra*  
lrwxrwxrwx 1 root root 15 10月 14 18:31 /dev/astrauvc -> bus/usb/001/009
```

This indicates that one port is down. This may be because a large program started at boot is using the camera. Close the terminal that started the large program. Test again by entering the command above:



```
jetson@yahboom:~$ ll /dev/astra*  
lrwxrwxrwx 1 root root 15 10月 16 09:59 /dev/astradepth → bus/usb/001/019  
lrwxrwxrwx 1 root root 15 10月 16 09:59 /dev/astrauvc → bus/usb/001/020  
jetson@yahboom:~$
```

If two ports are displayed, then modify the camera's device number to the one you just queried in the `run_docker.sh` script, and re-execute `run_docker.sh` to run the camera in Docker. It should then work normally.

If only one port is displayed even after closing the terminal that runs large programs, try unplugging and replugging the camera's connection to the main controller, then execute the command above again to check the port and test in the Docker container.